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BASIC VLAN CONFIGURATION LAB GUIDE

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CHAPTER 1: INTRODUCTION TO VLAN CONFIGURATION

1.1 Introduction to VLANs

This lab walks through building a small switched network from a blank configuration: creating VLANs, assigning switch ports to them, configuring a management VLAN, and enabling trunking between switches. Every command below is written out in full and labeled with the exact switch it belongs to, so you can follow along even if this is your first time configuring a Cisco switch.

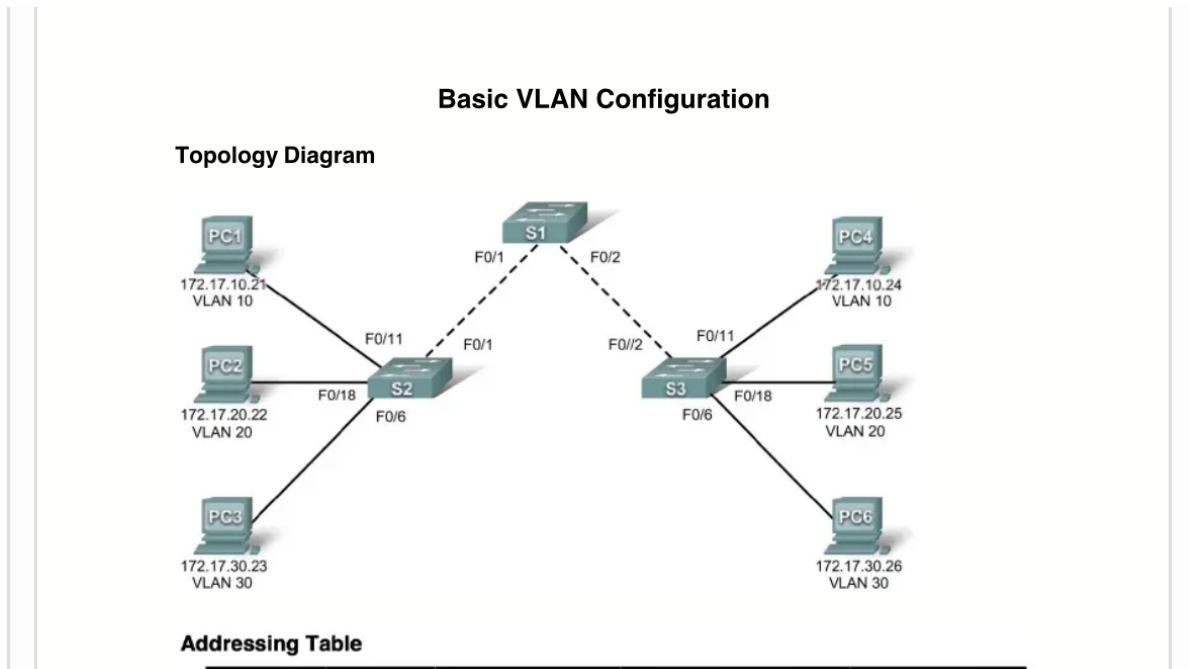


Figure 1: Topology Diagram

1.2 Topology Diagram

Three switches (S1, S2, S3) form the core of the network. S1 sits in the middle and trunks to S2 (via port Fa0/1) and to S3 (via port Fa0/2). Six PCs connect to S2 and S3, three per switch, one in each of the three data VLANs (10, 20, 30).

Reference Information

Keep this page handy - every IP address and port range used later in the lab comes from these two tables.

Addressing Table

Table 1: Addressing Table

Device (Hostname)	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 99	172.17.99.11	255.255.255.0	N/A

S2	VLAN 99	172.17.99.12	255.255.255.0	N/A
S3	VLAN 99	172.17.99.13	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	172.17.30.23	255.255.255.0	172.17.30.1
PC4	NIC	172.17.10.24	255.255.255.0	172.17.10.1
PC5	NIC	172.17.20.25	255.255.255.0	172.17.20.1
PC6	NIC	172.17.30.26	255.255.255.0	172.17.30.1

Initial Port Assignments (Switches S2 and S3)					
Table 2: Initial Port Assignments					
Ports		Assignment		Network	
Fa0/1 - 0/5		802.1Q Trunks (Native VLAN 99)		172.17.99.0/24	
Fa0/6 - 0/10		VLAN 30 - Guest (Default)		172.17.30.0/24	
Fa0/11 - 0/17		VLAN 10 - Faculty/Staff		172.17.10.0/24	
Fa0/18 - 0/24		VLAN 20 - Students		172.17.20.0/24	
Note: These same port ranges and VLAN numbers are reused in every task below. Whenever you see "Fa0/6-10" in a command, it refers to this table. Learning Objectives 3 Erase the startup configuration and reload a switch to its default (factory) state 3 Perform basic configuration tasks on a switch (hostname, passwords, DNS lookup) 3 Create VLANs 3 Assign switch ports to a VLAN 3 Add, move, and change ports between VLANs 3 Verify VLAN configuration 3 Enable trunking on inter-switch connections 3 Verify trunk configuration 3 Save the VLAN configuration					

TASK 1: PREPARE THE NETWORK

Step 1 - Cable the network Physically cable (or, in Packet Tracer, connect) the three switches and six PCs exactly as shown in the

their respective switch. Step 2 - Erase existing configuration and reload each switch Before configuring anything new, wipe out any leftover configuration so you are starting from a clean slate. Do this on all three switches individually - S1, S2, and S3 - by connecting to the console of each one and entering the following commands:

```
Switch>enable  
Switch#erase startup-config
```

Erasing the nvram filesystem will remove all configuration files! Continue? [confirm] [OK]
Erase of nvram: complete

```
Switch#reload
```

Proceed with reload? [confirm] Note: "erase startup-config" deletes the saved configuration file, and "reload" restarts the switch so it boots up with a blank (default) configuration. Do this for S1, then repeat for S2, then S3. After each switch reloads, it will prompt "Would you like to enter the initial configuration dialog?" - answer no so you land at the normal

```
Switch> prompt.
```

Step 3 - Initialize all ports to the shutdown state Still working one switch at a time, enter global configuration mode and administratively shut down every port. This gives you a known, consistent starting point before ports are re-enabled and assigned to VLANs later. Repeat this exact block on S1, S2, and S3:

```
Switch#configure terminal  
Switch(config)#interface range fa0/1-24  
Switch(config-if-range)#shutdown  
Switch(config-if-range)#interface range gi0/1-2  
Switch(config-if-range)#shutdown  
Switch(config-if-range)#end
```

Note: "Switch" here is a placeholder for whichever device you are currently configuring - you should see this same block run three times in your terminal history: once with the S1 prompt, once with S2, and once with S3.

TASK 2: PERFORM BASIC SWITCH CONFIGURATIONS

Step 1 - Configure hostname, passwords, and DNS lookup on each switch Apply the following guidelines to all three switches (S1, S2, S3), changing only the hostname each time. The example below shows the full set of commands for S1; repeat the same pattern for S2 and S3, substituting

the correct hostname:

Disable DNS lookup, so mistyped commands don't cause the switch to pause while trying to resolve them

as hostnames.

Set an EXEC (privileged) mode password of class.

Set a password of cisco for console (physical access) connections.

Set a password of cisco for vty (remote Telnet/SSH) connections.

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname S1
S1(config)#no ip domain-lookup
S1(config)#enable secret class
S1(config)#line console 0
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#exit
S1(config)#line vty 0 15
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#end
S1#copy running-config startup-config
```

Note: For S2, replace every "S1" in the block above with "S2"; for S3, replace it with "S3".

Everything else (the passwords and the DNS-lookup setting) stays the same on all three switches.

Step 2 - Re-enable the user-facing ports on S2 and S3 Step 3 of Task 1 shut down every port.

Now bring back only the ports that PCs are actually plugged into (Fa0/6,

Fa0/11, and Fa0/18 on both S2 and S3) and set them to access mode, meaning each port carries

single VLAN rather than acting as a trunk.

```
S2(config)#interface range fa0/6, fa0/11, fa0/18
S2(config-if-range)#switchport mode access
S2(config-if-range)#no shutdown
S2(config-if-range)#end
S3(config)#interface range fa0/6, fa0/11, fa0/18
S3(config-if-range)#switchport mode access
S3(config-if-range)#no shutdown
S3(config-if-range)#end
```

TASK 3: CONFIGURE AND ACTIVATE ETHERNET INTERFACES

Step 1 - Configure the PCs

On each PC, open its IP configuration settings (in Packet Tracer: click the PC Desktop tab IP Configuration) and manually enter the IP address, subnet mask, and default gateway from the

Addressing

Table on the reference page. You do not have to configure all six PCs to complete the lab. If you only want to test connectivity between two specific PCs - say PC1 and PC2 - it's enough to configure just those two with their addresses from the table. Configuring all six gives you the fullest picture and lets you test every VLAN, so it's the recommended approach if you have time.

TASK 4: CONFIGURE VLANS ON THE SWITCH

Step 1 - Create the VLANs on S1 From global configuration mode on S1, use the `vlan vlan-id` command to create each VLAN. This drops you into VLAN configuration mode, where the `name` command assigns a descriptive label. This lab uses four VLANs: 10 (faculty/staff), 20 (students), 30 (guest), and 99 (management - used for switch-to-switch traffic, not for end-user PCs).

```
S1(config)#vlan 10
S1(config-vlan)#name faculty/staff
S1(config-vlan)#vlan 20
S1(config-vlan)#name students
S1(config-vlan)#vlan 30
S1(config-vlan)#name guest
S1(config-vlan)#vlan 99
S1(config-vlan)#name management
S1(config-vlan)#end
S1#
```

Step 2 - Verify the VLANs were created on S1 The `show vlan brief` command lists every VLAN the switch knows about, its status, and which ports currently belong to it. Right after creation, all four new VLANs exist but have no ports assigned yet - every port is still sitting in the default VLAN 1.

```
S1#show vlan brief
VLAN Name Status Ports
```

1 default active

```
Fa0/1, Fa0/2, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9,
Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16,
Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23,
Fa0/24, Gi0/1, Gi0/2
```

10 faculty/staff active (none yet) 20 students active (none yet) 30 guest active (none yet) 99 management active (none yet) Step 3 - Create and name the same VLANs on S2 and S3 Repeat exactly the same set of commands from Step 1 - `vlan 10 / name faculty-staff`, `vlan 20 / name students`, `vlan 30 / name guest`, `vlan 99 / name management` - but this time on S2, and then again on S3. Confirm each with `show vlan brief` before moving on. What ports are currently assigned to the four VLANs you just created on S2 and S3?

Step 4 - Assign switch ports to VLANs on S2 and S3 Using the port-to-VLAN mapping from the Initial Port Assignments table on the reference page, assign each range of ports to its VLAN with the switchport access vlan vlan-id command. The interface range command lets you configure several ports at once instead of one at a time. The example below is written for S3 - run the identical block on S2 as well, since both switches use the same port layout.

```
S3(config)#interface range fa0/6-10
S3(config-if-range)#switchport access vlan 30
S3(config-if-range)#interface range fa0/11-17
S3(config-if-range)#switchport access vlan 10
S3(config-if-range)#interface range fa0/18-24
S3(config-if-range)#switchport access vlan 20
S3(config-if-range)#end
S3#copy running-config startup-config
```

Destination filename [startup-config]? [enter] Building configuration... [OK] Note: Don't forget the final "copy running-config startup-config" step - this saves the configuration to memory so it survives a reboot. Run it on S3, then repeat the whole block (commands and save) on S2. Step 5 - Confirm which ports were added On S2, run show vlan id vlan-number to see only the single VLAN you specify, which is easier to read than the full show vlan brief output when you just want to check one VLAN.

```
S2#show vlan id 10
```

Which ports are assigned to VLAN 10 on S2?

Note: "show vlan id vlan-name" (using the name instead of the number, e.g. faculty/staff) produces the same

output. You can also check a single port's VLAN membership with "show interfaces interface switchport" - for

example, "show interfaces fa0/6 switchport" on S2. Step 6 - Assign the management VLAN IP address A management VLAN is the VLAN you use to remotely reach and administer the switch itself (over HTTP, Telnet, SSH, or SNMP) - separate from the VLANs that carry regular user traffic. By default a switch uses VLAN 1 for this, but VLAN 1 is a poor choice because every port defaults to it, so anyone plugging into any unconfigured port could reach the switch's management interface. This lab uses VLAN 99 instead, which was created in Step 1. Give each switch's VLAN 99 interface an IP address from the Addressing Table. Run the matching block on

each of the three switches:

```
S1(config)#interface vlan 99
S1(config-if)#ip address 172.17.99.11 255.255.255.0
S1(config-if)#no shutdown
S1(config-if)#end
S2(config)#interface vlan 99
S2(config-if)#ip address 172.17.99.12 255.255.255.0
S2(config-if)#no shutdown
S2(config-if)#end
S3(config)#interface vlan 99
S3(config-if)#ip address 172.17.99.13 255.255.255.0
S3(config-if)#no shutdown
S3(config-if)#end
```

Note: Assigning this address does two things: it lets the three switches ping and manage each other, and it lets any host physically connected to a port assigned to VLAN 99 reach the switches too. Because of that second point, VLAN 99 ports should be treated as sensitive and restricted to trusted devices only. Step 7 - Configure trunking and the native VLAN on all switches A trunk port is a link between two switches that carries traffic for every VLAN at once, instead of being restricted to a single VLAN the way an access port is. Trunks are what allow, for example, a PC in VLAN 10 on S2 to reach a PC in VLAN 10 on S3 through S1. Because this lab uses 2960-series switches, which only understand the 802.1Q trunking standard, there's no need to explicitly choose an encapsulation type - 802.1Q is the only option. Every 802.1Q trunk also has a native VLAN: the one VLAN whose traffic is sent untagged (without an 802.1Q tag) across the link. This lab uses VLAN 99 as the native VLAN on every trunk. It is best practice to use a VLAN other than the default VLAN 1 for this, so that unlabeled traffic doesn't automatically fall into the same VLAN every device defaults to. Run the following on S1, S2, and S3 to turn the inter-switch ports (Fa0/1 through Fa0/5 on each switch) into trunks with native VLAN 99:

```

S1(config)#interface range fa0/1-5
S1(config-if-range)#switchport mode trunk
S1(config-if-range)#switchport trunk native vlan 99
S1(config-if-range)#no shutdown
S1(config-if-range)#end
S2(config)#interface range fa0/1-5
S2(config-if-range)#switchport mode trunk
S2(config-if-range)#switchport trunk native vlan 99
S2(config-if-range)#no shutdown
S2(config-if-range)#end
S3(config)#interface range fa0/1-5
S3(config-if-range)#switchport mode trunk
S3(config-if-range)#switchport trunk native vlan 99
S3(config-if-range)#no shutdown
S3(config-if-range)#end

```

Step 8 - Verify the trunks On S1, confirm both trunk links are up and using the settings you just configured:

```

S1#show interface trunk
Port Mode Encapsulation Status Native VLAN
Fa0/1 on 802.1q trunking 99
Fa0/2 on 802.1q trunking 99
Port VLANs allowed on trunk
Fa0/1 1-4094
Fa0/2 1-4094
Port VLANs allowed and active in management domain
Fa0/1 1,10,20,30,99
Fa0/2 1,10,20,30,99
Port VLANs in spanning tree forwarding state and not pruned
Fa0/1 1,10,20,30,99
Fa0/2 1,10,20,30,99

```

TASK 5: VERIFY CONNECTIVITY

Step 1 - Ping between switches From S1, ping the VLAN 99 management address of both S2 and S3 to confirm the trunks and management VLAN are working end to end.

```

S1#ping 172.17.99.12
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.99.12, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/9 ms
S1#ping 172.17.99.13
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.99.13, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms

```

Note: A single dropped ping (the leading "." before the four "!" marks) on the very first attempt to a new destination is normal - it usually reflects the short delay while the switch resolves the destination's MAC address via ARP, not a real connectivity problem. Step 2 - Ping several hosts from PC2 Open PC2's command prompt (Desktop tab Command Prompt in Packet Tracer) and try each ping below. PC2 is in VLAN 20; think about which of these destinations shares that VLAN before you predict the result. Ping from PC2 to PC1 (172.17.10.21). Is the ping attempt successful?

Ping from PC2 to the S2 management address, 172.17.99.12. Is the ping attempt successful?

Ping from PC2 to PC5 (same VLAN, different switch). Is the ping attempt successful?

Step 3 - Move PC1 into the same VLAN as PC2 PC1 is currently wired into a port on S2 that's assigned to VLAN 10. To place PC1 in the same VLAN as PC2 (VLAN 20), change the VLAN membership of the switch port PC1 is connected to - not PC1's own IP settings - using switchport access vlan on that interface. Which command(s) would you use on S2 to move PC1's port into VLAN 20?

After making that change, ping from PC2 to PC1. Is the ping attempt successful? Why?

Step 4 - Change PC1's IP address and network Now that PC1's switch port lives in VLAN 20, PC1 also needs an IP address that matches VLAN 20's network (172.17.20.0 /24) instead of its original VLAN 10 address - devices can only communicate directly when they're on the same IP subnet as well as the same VLAN. After re-addressing PC1 into the 172.17.20.0/24 network, which other device(s) can PC2 now reach?

APPENDICES

Appendix A: